```
except-interface eth0
except-interface ppp0
```

2. Set DNSMASQ to reference */etc/resolv.upstream* to get the list of upstream name servers:

```
resolv-file=/etc/dnsmasq/resolv.upstream
```

3. Increase the cache size to 1024:

```
cache-size=1024
```

4. Set the domain name to anything you like:

```
domain=anything-you-like.local
```

5. Now the main part: enable DHCP and set the lease time to 'infinite':

```
dhcp-range=192.168.1.10,192.168.1.254,255.255.255.0,infinite
```

6. Set up a Microsoft Windows hack to release DHCP leases on shutdown:

```
dhcp-option=vendor:MSFT,2,1i
```

7. Set up the PXE server:

```
dhcp-boot=pxelinux.0
enable-tftp
tftp-root=/tftpboot
```

8. Set some miscellaneous options:

```
dhcp-leasefile=/etc/dnsmasq/dnsmasq.leases
dhcp-script=/bin/echo
log-queries
log-dhcp
```

9. Disable authoritative DHCP: Comment the line, that is put a # sign at the beginning of the line that says "dhcp-authoritative".

Right now, the configuration is a bit broken. Create the directories */tftpboot* and */tftpboot/pxelinux.cfg*.

Download the latest *syslinux tar.bz2* file from *www.kernel.org/pub/linux/utils/boot/syslinux/Testing/*. Don't worry, 'Testing' releases aren't that broken. Extract it and copy the files *core/pxelinux.0* and *com32/menu/menu.c32 to /tftpboot*.

As an exercise, we are going to download SystemRescueCD and set it up to boot over the network. Download the latest ISO from *nchc.dl.sourceforge.net/sourceforge/systemrescuecd/systemrescuecd-x86-1.1.4.iso*, mount it and copy files *rescuecd*, *initram.igz*, *sysrcd.dat*, and *sysrcd.md5* to */tftpboot/sysrcd*.

Copy the following lines to the */tftpboot/pxelinux.cfg/default* file:

```
default menu.c32
prompt 0
menu title PXE Boot Menu

label sysrcd
menu label Boot SystemRescueCD
kernel sysrcd/rescuecd
append initrd=sysrcd/initram.igz setkmap=us vga=791 boottftp=tftp://19
2.168.1.1/sysrcd/sysrcd.dat
```

Disable PEERDNS if you are using PPPOECONF by commenting (#) the "usepeerdns" line in the */etc/ppp/peers/dsl-provider* file.

Now restart the dnsmasq server:

```
# /etc/init.d/dnsmasq restart
```

That should do it. Any machine that accesses PXE when booting will boot into the PXE Bootloader installed on your server. Find out from the motherboard manual of your client how to boot from the network. When you do so, you will be presented with a menu with a single entry.

Press *Enter* now. The kernels, rootfs and the datfile will be downloaded through TFTP and the kernel will be executed. Thus SysRCD will boot up on the client. It will get all its network configuration from DNSMASQ (DHCP).

What next?

Try browsing the Internet from your newly PXE-booted PC. If all goes well, you should be able to.

Tips: You don't need to shut down or restart your server at all from now on. Keep it as it is, and try setting a Guinness Book World Record. On a more serious note, all maintenance, installations and removals will be performed dynamically.

Coming up next, we'll set up two Web server instances on the same server: one to serve your Intranet and one to serve the public through the Internet. Till then, see you! END

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